

Omar Ayman Mohamed

Cairo, Egypt | +20 1009398538 | omarayman40404@gmail.com
linkedin.com/in/omar-ayman-723b401b4 | github.com/OmarAyman415

SUMMARY

Computer Science graduate with hands-on experience in .NET development and backend engineering. Proficient in building REST APIs and microservices, with exposure to relational and non-relational databases, containerization, and distributed messaging. Holds a strong foundation in OOP, SOLID principles, and design patterns, with a consistent record of delivering complete, production-ready projects.

TECHNICAL SKILLS

Core .NET Stack: C#, ASP.NET MVC, ASP.NET Core, .NET Framework, Entity Framework / EF Core

Other Languages: Java, JavaScript, Python, Dart, PHP, C++, SQL

Frameworks: Spring Boot, Spring Security, Express.js, Node.js, React, Flutter, JavaFX

Databases: SQL Server, PostgreSQL, MySQL, MongoDB, SQLite

Tools: Docker, Apache Kafka, Git, GitHub, JPA, Hibernate, Drift ORM, WebSockets

Concepts: RESTful APIs, Microservices, OOP, SOLID Principles, Design Patterns, DSA, JWT Auth, MVC

WORK EXPERIENCE

IT Operations Engineer

May 2025 – Present

e& Egypt (Etisalat Egypt)

Cairo, Egypt

- Maintain IT infrastructure, monitor system performance, and ensure high availability of services at one of Egypt's largest telecoms.

PROJECTS

Instagram Clone / C#, ASP.NET MVC, Entity Framework, Bootstrap, SQL Server

- Full-featured photo and video sharing platform with private messaging and user authentication.
- Built with ASP.NET MVC and Entity Framework on a SQL Server backend, with a responsive Bootstrap frontend.

Compiler Scanner / C#, ASP.NET, .NET

- Lexical scanner for a case-sensitive OOP language with a .NET-based UI.
- Demonstrates deep familiarity with the .NET platform and language processing concepts.

Real-Time Chat Application / Java, Spring Boot, WebSockets, JWT, Spring Security

- Secure real-time messaging using WebSockets and JWT/Spring Security authentication.
- Architecture mirrors ASP.NET Core + SignalR patterns; easily portable to the .NET stack.

E-Commerce REST API / Java, Spring Boot, PostgreSQL, JPA/Hibernate

- Full REST API backend handling products, carts, and orders via JPA/Hibernate.
- Comparable in design to an ASP.NET Core Web API project with Entity Framework Core.

Microservices Architecture App / Node.js, Express.js, Apache Kafka, Docker

- Async inter-service communication via Apache Kafka, fully containerized with Docker.

Lift Tracker / Flutter, Dart, SQLite, Drift ORM

- Cross-platform workout tracking app with an offline-first SQLite database via Drift ORM.

Stereo R-CNN 3D Object Detection / Python, Deep Learning, Computer Vision

- LiDAR-free 3D object detection system for autonomous driving using stereo cameras.
- Validated on the KITTI benchmark with competitive accuracy.

Job Board REST API / C#, ASP.NET Core Web API, Entity Framework Core, SQL Server, JWT

- Built a fully documented REST API for job listings, applications, and employer/candidate profiles.
- Implemented JWT-based authentication with role-based authorization (Admin, Employer, Candidate) using ASP.NET Core Identity.
- Used EF Core with SQL Server, repository pattern, and pagination for scalable data access.

Fingerprint Recognition System / Python, ANN

- ANN-based biometric identification analyzing ridge endings and bifurcations.

Course Management System / Java, JavaFX

- Desktop MVC app for managing courses, students, and assignments.

EDUCATION

Bachelor of Science in Computer Science

Aug 2019 – Jun 2023

Helwan University, Faculty of Computer and Artificial Intelligence

GPA: 3.4 / 4.0

LANGUAGES

Arabic: Native | **English:** Intermediate